YadYar Lite: Early Prediction of At-Risk Stu(

Phase 2 Final Report - Topic T-29

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Presentation order: Atena Molaee -> Fatemeh Masoomi -> Mahdieh Tajbakhsh

Executive Summary

This project investigates whether students who later fail or withdraw can be identified using only information available during the first 30 days of an online course. The study uses a small, reproducible subset of OULAD (module AAA, presentation 2013J) and focuses on class imbalance, minority-class recall, threshold selection, and asymmetric error cost. The output is an early-warning signal for supportive human review, not an automatic academic decision.

Dataset	**Primary baseline**	**Selected high-recall point**	**Bonus comparison**
383 students 27.4% At Risk	Logistic Regression	Balanced Logistic threshold 0.30 Recall 0.857 Cost 45	LightGBM Best PR-AUC 0.654

Research Question

Can students who will later fail or withdraw be identified using demographic, enrollment, assessment, and VLE activity information observed during course days 0-30?

Main Finding

Accuracy alone is misleading. A majority classifier reaches 72.7% Accuracy while finding no at-risk students. Balanced Logistic Regression at threshold 0.30 identifies 18 of 21 at-risk students and minimizes the illustrative 5:1 error cost, although it generates more false alerts. Standard LightGBM is a useful conservative alternative with 71.4% Precision and only four false positives.

Project Contribution

- A reproducible first-30-day student-risk classification pipeline.
- An explicit comparison of standard and class-balanced learning.
- Operating-point analysis at thresholds 0.50 and 0.30.
- Representative error analysis, result breakdown, and a lightweight demo.

1. Dataset, Target, and Features

The Open University Learning Analytics Dataset (OULAD) contains linked student, course, assessment, and Virtual Learning Environment tables. To keep the project lightweight, the experiment uses only module AAA, presentation 2013J.

Class	Definition	Students	Share
Not At Risk (0)	Pass or Distinction	278	72.6%
At Risk (1)	Fail or Withdrawn	105	27.4%

Features

Numerical features include previous attempts, studied credits, total clicks, active days, unique resources, clicks per active day, early assessment count, and early average score. Categorical features include gender, region, prior education, deprivation band, age band, disability status, module, and presentation.

Leakage Prevention

- `final_result` is used only to create the target and is excluded from model inputs.
- VLE and assessment events are restricted to days 0-30.
- Preprocessing is fitted only on the training split.

2. Models and Experimental Setup

Mandatory models are a majority Dummy classifier, standard Logistic Regression, and class-balanced Logistic Regression. As optional bonus work, standard and class-balanced LightGBM classifiers are included. LightGBM complexity is deliberately limited to 100 trees, learning rate 0.05, maximum depth 3, seven leaves, and L2 regularization.

- Split: 80% train / 20% test, stratified, random_state=42.
- Test set: 77 students (56 Not At Risk, 21 At Risk).
- Decision thresholds: 0.50 and 0.30.
- Illustrative error cost: Total Cost = 5 x FN + 1 x FP.

3. Evaluation Metrics

The primary metrics are At-Risk Recall and PR-AUC. Recall measures how many truly at-risk students are found. PR-AUC evaluates minority-class ranking across thresholds. Precision, F1, Accuracy, ROC-AUC, Brier score, confusion matrices, and asymmetric cost provide supporting evidence.

4. Baseline and Model-Comparison Results

Model	Threshold	Accuracy	Precision	Recall	F1	PR-AUC	Brier	FP	FN	Cost
Dummy Majority	0.50	0.727	0.000	0.000	0.000	0.273	0.273	0	21	105
Standard Logistic	0.50	0.766	0.600	0.429	0.500	0.587	0.171	6	12	66
Standard Logistic	0.30	0.636	0.370	0.476	0.417	0.587	0.171	17	11	72
Balanced Logistic	0.50	0.649	0.393	0.524	0.449	0.596	0.202	17	10	67
Balanced Logistic	0.30	0.571	0.375	0.857	0.522	0.596	0.202	30	3	45
Standard LightGBM	0.50	0.805	0.714	0.476	0.571	0.644	0.160	4	11	59
Standard LightGBM	0.30	0.714	0.478	0.524	0.500	0.644	0.160	12	10	62
Balanced LightGBM	0.50	0.675	0.423	0.524	0.468	0.654	0.196	15	10	65
Balanced LightGBM	0.30	0.584	0.378	0.810	0.515	0.654	0.196	28	4	48

Interpretation

- The Dummy classifier shows majority-class collapse: 72.7% Accuracy and 0% At-Risk Recall.
- Balanced Logistic at threshold 0.30 is the selected high-recall operating point: Recall 0.857, three false negatives, and the lowest cost (45).
- Standard LightGBM at threshold 0.50 is the strongest conservative point: Accuracy 0.805, Precision 0.714, F1 0.571, Brier 0.160, and four false positives.
- Balanced LightGBM achieves the best PR-AUC (0.654); at threshold 0.30 its Recall is 0.810 and cost is 48.

5. Operating-Point Analysis

The preferred model depends on the institution's operational goal. Lowering the threshold increases Recall but also creates more alerts. The project prioritizes avoiding missed at-risk students, so Balanced Logistic at threshold 0.30 is used in the demo. Standard LightGBM at threshold 0.50 remains a defensible option when tutor capacity is limited and unnecessary alerts must be minimized.

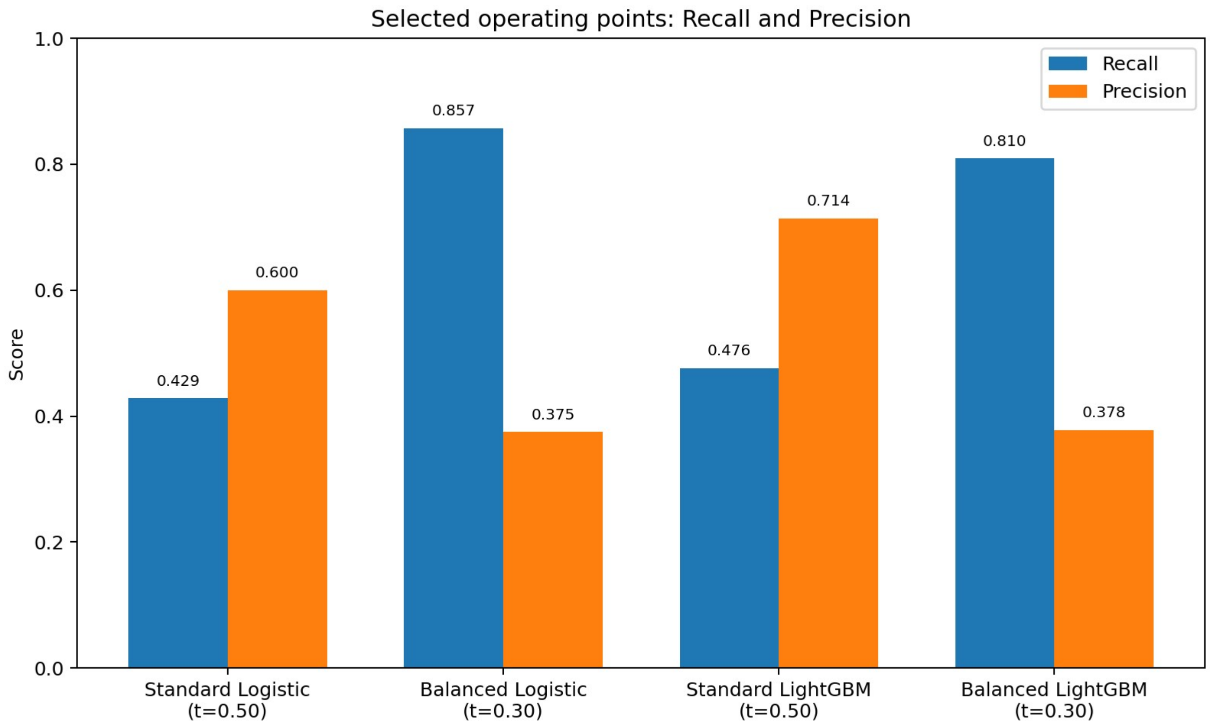

Figure 1. Recall-precision trade-off at selected operating points.

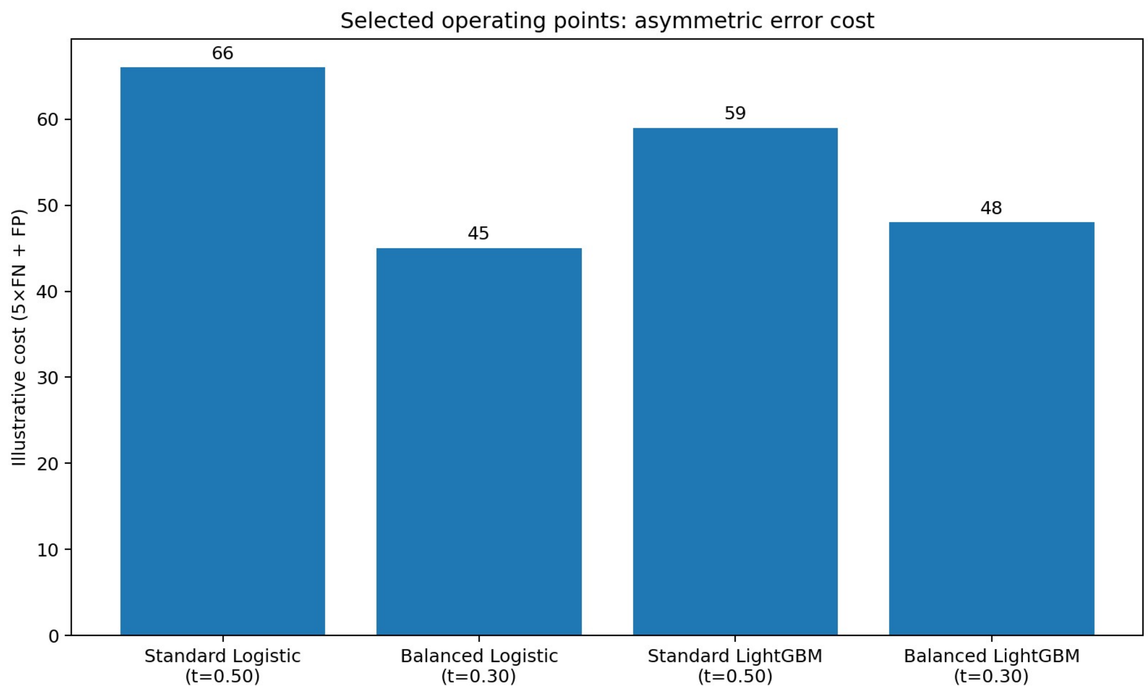

Figure 2. Illustrative asymmetric cost, where a false negative costs five times a false positive.

6. Precision-Recall Behaviour and Feature Importance

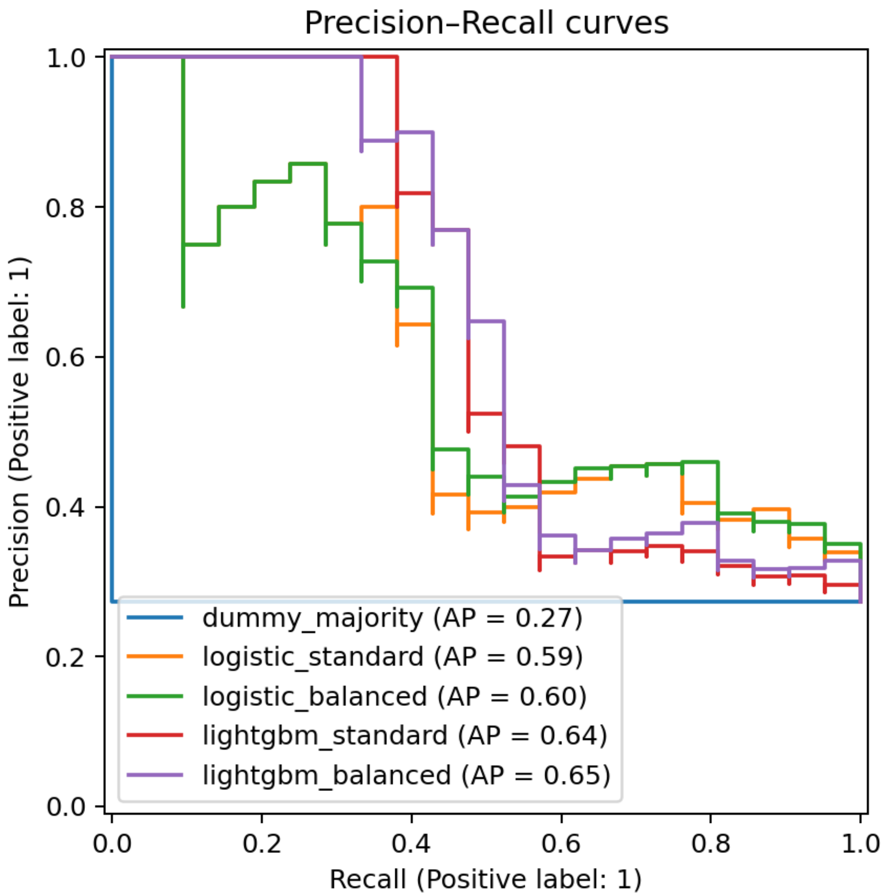

Figure 3. Precision-recall curves for the evaluated models.

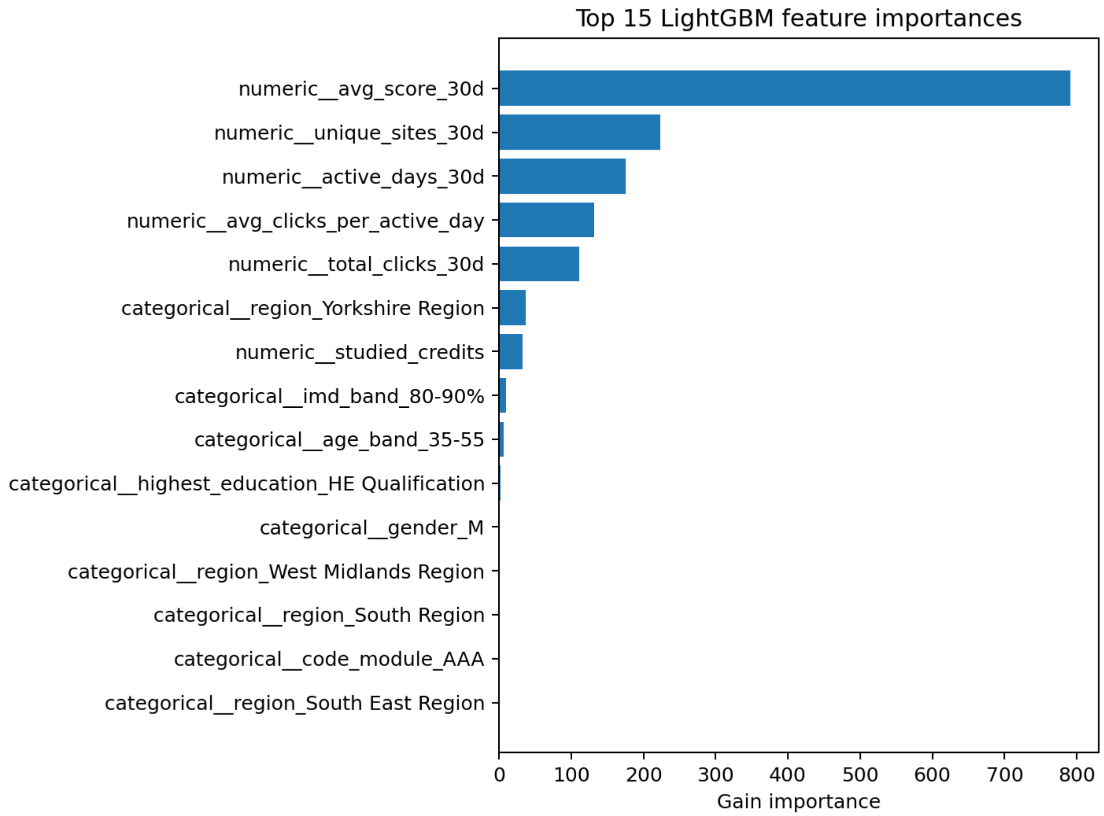

Figure 4. LightGBM split-based feature importance.

The most important LightGBM signal is early average assessment score, followed by unique resources, active days, average clicks per active day, and total clicks. These importances describe the fitted model and must not be interpreted as causal effects.

7. Result Breakdown by Early Engagement

Engagement group	Students	At-risk rate	Recall	Precision
Low	26	46.2%	91.7%	52.4%
Medium	25	24.0%	66.7%	30.8%
High	26	11.5%	100.0%	21.4%

At-risk students are most common in the Low group. Recall is weakest in the Medium group. The High group has 100% Recall but includes only three at-risk students, so this result should not be generalized.

8. Representative Error Analysis

False Negatives

Student	Outcome	Probability	Group	Clicks	Active days	Early score
185439	Fail	0.297	Low	200	8	74
603861	Withdrawn	0.269	Medium	236	17	61
155984	Withdrawn	0.192	Medium	278	10	85

One false negative is very close to the decision threshold. The two withdrawn students show that apparently reasonable early engagement and scores cannot always reveal later withdrawal.

False Positives

The selected high-recall model produces 30 false positives. Typical cases have weak or missing early assessment evidence but later recover and pass. Therefore, a flag should initiate supportive human review rather than label a student as certain to fail.

9. Lightweight Demo and Integration Contract

The project contains a console demo and a Streamlit interface. The saved demo model is Balanced Logistic Regression at threshold 0.30.

```
{
  "student_id": "optional_external_id",
  "observation_window_days": 30,
  "risk_probability": 0.74,
  "decision_threshold": 0.30,
  "at_risk_flag": true,
  "recommended_action": "supportive_human_review"
}
```

A future learning assistant could send this object to a tutor dashboard. Staff would inspect recent activity and offer support. The result must not be used to punish, restrict, or permanently label a student.

10. Limitations

- Only one course presentation and 383 students are studied; the test set contains 21 at-risk students.
- A random split does not simulate deployment on a later course presentation.
- The 5:1 cost ratio and threshold 0.30 are illustrative and should reflect real support capacity.
- The high-recall model flags 48 of 77 test students, which may be operationally expensive.
- Missing assessment evidence and an actual zero score are not fully distinguished.
- Events after day 30 cannot be observed, and demographic variables require fairness review.
- LightGBM importance is model-specific rather than causal; extensive tuning was intentionally avoided.

11. Future Work

- Separate not submitted, not yet scored, and actual zero-score cases.
- Train on an earlier presentation and test on a later presentation.
- Choose thresholds based on actual tutor capacity and real error costs.
- Evaluate calibration and Recall across demographic subgroups.

12. Conclusion

Accuracy alone is unsuitable for this imbalanced early-warning task. LightGBM improves ranking, precision, and probability quality, but Balanced Logistic Regression at threshold 0.30 remains the selected high-recall point because it misses only three at-risk students and has the lowest cost under the assumed 5:1 ratio. The two models therefore support different operating needs: high Recall versus fewer unnecessary alerts.

Reproducibility

Install dependencies with `pip install -r requirements.txt`, run the pipeline with `run_phase2.bat` or `./run_phase2.sh`, run the console demo with `python src/predict_examples.py`, and start the interface with `streamlit run app.py`. The project includes the processed data, generated outputs, and saved model.

References and AI Acknowledgement

Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2015). Open University Learning Analytics Dataset. UCI Machine Learning Repository. DOI: 10.24432/C5KK69.

Ke, G., et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. Advances in Neural Information Processing Systems.

Generative AI assistance was used for code organization, debugging, documentation drafting, and explanation. All experiments were executed and checked, and the team is responsible for understanding and defending the submitted work.

Optional Bonus Work

The core Phase 2 requirements are complete before the following optional extensions. These experiments are reported separately so the baseline study remains clear and lightweight.

Extension	What we added	Credit
Additional dataset slice	Second breakdown by early-assessment submission status (separate from the required engagement breakdown).	+2
Simple model comparison	Logistic Regression baseline compared with small Standard and Balanced LightGBM models.	+3
Lightweight improvement attempt	Added an explicit has_early_assessment feature. The held-out metrics did not improve, so the simpler feature set was retained.	+3
Cleaner demo package	README, reproducible example inputs, expected outputs, console demo and Streamlit demo.	+2

Potential bonus coverage: +10 points, subject to the course staff's grading decision.

Additional Slice Result

At threshold 0.30, 8 test students had no early assessment submission; 7 were actually at risk and all 7 were detected (Recall = 1.00), with one false positive. Among 69 students who submitted an early assessment, Recall = 0.786. The first slice is small, so the result is descriptive rather than generalizable.

Feature Improvement Attempt

A binary has_early_assessment feature was added to distinguish non-submission from a numeric score of zero. Accuracy, Precision, Recall, F1, FN, and FP were unchanged; PR-AUC changed from 0.5963 to 0.5960. Because the extra feature was redundant with assessment_count_30d, the simpler model was retained.

Reproducibility: run python src/bonus_analysis.py. Detailed evidence is included in BONUS_WORK.md and the bonus output CSV/PNG files.